

ISHAN MADHURANGA

Software Engineer

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WORK EXPERIENCE

Software Engineer

Sysco LABS

Oct 2024 – Present Colombo, Sri Lanka

- Contributing to the development of a data integration and onboarding platform to incorporate data from acquired companies into internal systems.
- Collaboration with the team to build dashboards that support onboarding workflows.
- Supporting the development of workflow-driven interfaces for validation and onboarding actions.
- Technologies - React.js | Python | AWS | Jenkins**

Software Engineer (AI/ML) - Intern

IronOne Technologies

Aug 2023 – Jan 2024 Colombo, Sri Lanka

- Developed and optimized time series machine learning models using deep learning, hyperparameter tuning, and ensemble methods.
- Automated and streamlined model training pipelines for efficient, scalable cloud deployment.
- Researched novel algorithms and implemented model explainability using SHAP to improve accuracy and transparency.
- Technologies - Python | TensorFlow | Keras | Snowflake | AWS SageMaker**

PUBLICATIONS

Machine Learning Based Enhanced Remote Power Analysis Attacks

5th International Conference on Advanced Research in Computing (ICARC 2025)

Jan 2024 – Feb 2025 Presented at ICARC 2025

GitHub | Project Page

- Built deep learning models (CNN, LSTM) to analyze power trace data for cryptographic key extraction.
- Applied machine learning techniques to improve the accuracy and success rate of Remote Power Analysis attacks.
- Implemented preprocessing and model training pipelines for sensor trace analysis.
- Published on IEEE Xplore. DOI: 10.1109/ICARC64760.2025.10963016**
- Contribution – Developed and implemented the deep learning-based attack model.**

EDUCATION

B.Sc.(Hons) in Computer Engineering

University of Peradeniya | CGPA : (3.65/4.00)

Jan 2020 – Aug 2024

TECHNICAL SKILLS

JavaScript	●●●●●●
React.js	●●●●●●
Python	●●●●●●
Node.js	●●●●●●
Java	●●●●●●
C	●●●●●●
PostgreSQL	●●●●●●
MySQL	●●●●●●
MongoDB	●●●●●●
Tensorflow / Keras	●●●●●●
AWS	●●●●●●
Git / GitHub	●●●●●●

STRENGTHS

- Problem Solving
- Hard Working
- Team Collaboration
- Adaptability
- Time Management

CERTIFICATIONS

- Java Object-Oriented Programming
- Java: Data Structures
- Programming for Everybody (Getting Started with Python)
- Machine Learning Specialization
- Neural Networks and Deep Learning
- Structuring Machine Learning Projects

PROJECTS

Smart Safety Helmet for Large Scale Construction Sites

Group

📅 Oct 2022 – Jan 2023

🔗 [GitHub](#) | 📄 [Project Page](#) | 🌐 [Website](#)

- An embedded system project used to supervise the workers of a construction site.
 - Contains a web-based application for the manager and a mobile app for the workers.
 - danger zones will be highlighted on the site map in different colors.
 - **Contribution - Entire web application.**
 - **Technologies - React.js | Firebase | Node.js | Flutter | Postman | Vercel**
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Postgraduate Student Management System

Group

📅 Nov 2021 – Mar 2022

🔗 [GitHub](#) | 📄 [Project Page](#)

- A web-based application to keep track of the academic program of postgraduate students of the Department of Computer Engineering.
 - Students will be notified two weeks before the deadline.
 - **Contribution - Implemented Automatic Event Management System and developed some backend functionalities.**
 - **Technologies - MongoDB | Express.js | React.js | Node.js | Postman**
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Automatic Email Sender

Individual

📅 Jul 2022 – Aug 2022

🔗 [GitHub](#)

- Code is written in ES6 for modern JavaScript support.
 - Implemented functionality to send emails to a given list of recipients.
 - Supported templated emails for customization.
 - Added scheduling capabilities to send emails at specified times.
 - **Technologies - Node.js**
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Student Result Management System

Group

📅 Nov 2021 – Mar 2022

🔗 [GitHub](#)

- An android application developed for undergraduates to keep track of their academic performance.
- Students can calculate the average grade required for the expected degree class for the upcoming semesters.
- **Contribution - built all the functionalities of the app.**
- **Technologies - Java | Android Studio | MySQL | PHP**

REFEREES

Lushanthan Sivaneasharajah

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✉ Head of Data Science

IronOne AI Labs

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